

BLOOD MANAGEMENT SOFTWARE

DETAILED FEATURES FOR EACH DASHBOARD

Problem Statement

Currently in India

- Hospitals struggle to find required blood in emergencies
- No centralized system to check real-time blood availability
- Doctors manually call multiple blood banks
- Transportation of blood is unorganized
- Delay leads to loss of patient life

As highlighted in your concept

Families and doctors search multiple places → delay → critical risk

Proposed Solution

We will build a Multi-User Blood Management System with:

Core Modules:

1. Hospital / Doctor Dashboard
2. Blood Bank / Admin Dashboard
3. Rider / Logistics App
4. Public Landing Website (for camps & awareness)

User Roles (Multi Login System)

Role

Access

Admin (Super User)	Full control
Blood Bank	Manage blood & requests
Hospital / Doctor	Request blood
Rider	Pickup & delivery
Donor (optional future)	Register & track

HOSPITAL / DOCTOR DASHBOARD FEATURES (Core Features)

1. Login & Authentication

- Secure login (OTP / Password)

2. Search Blood Availability

Select

- Blood Group (A+, B-, etc.)
- Quantity

3. System shows:

- Nearest blood banks
- Available stock

4. Raise Blood Request

- Send request to selected blood bank
- ADD:-
 - Patient details
 - Urgency level

5. Request Tracking

Status:

- Pending
- Accepted
- Rejected
- In Delivery

6. Notifications

- Blood bank accept/reject alert

7. History Dashboard

- Monthly / yearly blood usage
- Reports & analytics

8. Emergency Mode

- Highlight rare/urgent requests

BLOOD BANK / ADMIN DASHBOARD

Admin (Single Super User)

- **Add / Manage:**
 - Blood banks
 - Hospitals
 - Riders
- System monitoring

Blood Bank Features

1. Blood Inventory Management

- Add blood units
 - Blood group
 - Quantity
 - Collection date
 - Expiry date
- Categorization
 - PCV, FFP, SDP

2. Real-Time Stock Update

- Auto update on:
 - Request approval
 - Delivery

3. Request Management

- View hospital requests
- Accept / Reject
- Assign rider

4. Donor Management

- Store donor data:
 - Name
 - Blood group
 - Donation history

5. Blood Camp Management

- **Create campaign**
- **Track:**
 - Location
 - Date
 - Total units collected

6. Reports & Analytics

- Monthly collection
- Supply reports
- Hospital-wise data

7. Logistics Dashboard

- Track riders
- Delivery status
- ETA

RIDER / DELIVERY APP FEATURES

Core Features

1. Login (Mobile App)

2. Task Assignment

- Pickup request from blood bank

3. Navigation

- Map integration (Google Maps)

4. Pickup Workflow

Collect:

- Blood unit
- Documents

5. Delivery Workflow

- Deliver to hospital
- Confirmation (OTP / Signature)

6. Live Tracking

- Real-time location tracking

7. Status Update

- Picked
- In Transit
- Delivered



END-TO-END SYSTEM FLOW

Complete Flow:

Step 1: Blood Collection

- Donor → Camp → Blood Bank storage

Step 2: Inventory Update

- Blood bank updates stock in system

Step 3: Hospital Request

Hospital:

- Select blood group
- Enter quantity
- Submit request

Step 4: Smart Matching

System:

- Finds nearest blood banks
- Filters based on:
 - Availability
 - Distance

Step 5: Blood Bank Action

- Accept / Reject request

Step 6: Rider Assigned

- Blood bank assigns rider

Step 7: Pickup & Testing

- Rider:
 - Collects sample (if needed)
 - Documentation

Step 8: Delivery

- Blood transported using cold storage
- Cold chain requirement

Step 9: Hospital Receives

- Confirm delivery

Step 10: System Update

- Inventory updated
- Reports generated

TECH STACK

Frontend

- React.js
- Tailwind CSS

Backend

- JAVA SpringBoot

Database

- MYSQL

Mobile App

- Flutter / React Native (Rider app)

APIs & Services

- Google Maps API (location)
- Firebase (notifications)
- Socket.io (real-time updates)

Cloud

- AWS / Azure / DigitalOcean

SECURITY FEATURES

- Role-based access control
- Data encryption
- Secure API authentication (JWT)
- Audit logs